

Section : Biology

Q.1 Which one of the followings would increase population density after certain time?

1. Natality and mortality
2. Mortality and emigration
3. Natality and emigration
4. Natality and immigration

Options 1. 1

2. 2
3. 3
4. 4

Q.2 *Drosophila melanogaster* was chosen by Thomas Hunt Morgan for the experimental verification of the chromosomal theory of inheritance because of its:

- (A) Many types of hereditary variations visible under a low-powered microscope
- (B) Large number of progeny in single mating
- (C) Clear differentiation of sexes
- (D) Life cycle of about 4 weeks

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.3 A typical angiosperm anther is generally:

1. Single lobed
2. Bilobed
3. Trilobed
4. Spherical

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Identify the correct sequence of occurrence of geological period from recent to oldest time?

- (A) Permian
- (B) Jurassic
- (C) Silurian
- (D) Devonian

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (C), (B), (A)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Which of the followings are the side effects of the usage of anabolic steroids in females?

- (A) Deepening of voice
- (B) Increased aggressiveness
- (C) Mood swings
- (D) Abnormal menstrual cycle

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Some strains of *Bacillus thuringiensis* produce toxic proteins that do not kill:

1. Lepidopterans
2. Coleopterans
3. Crustaceans
4. Dipterans

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Identify the primary lymphoid organs, where immature lymphocytes differentiate into antigen sensitive lymphocytes.

- (A) Bone marrow
- (B) Spleen
- (C) Thymus
- (D) Appendix

Choose the correct answer from the options given below:

1. (B) and (D) only
2. (A) and (C) only
3. (A), (B) and (D) only
4. (A), (B), (C) and (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.8 Which one of the following is not the autosome-linked recessive Mendelian disorder?

1. Sickle-cell anaemia
2. Haemophilia
3. Phenylketonuria
4. Thalassemia

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Which one of the following statements is not true with reference to AIDS infection in human beings?

1. It is caused by retro viruses.
2. There is always a time lag between the infection and the appearance of symptoms.
3. AIDS does not spread by shaking hands with an infected person.
4. Infected cell can not survive while viruses are being replicated and released.

Options 1. 1

2. 2
3. 3
4. 4

Q.10 The typical angiosperm embryo sac at maturity has:

1. Two synergids, one egg cell, three antipodals
2. Two synergids, one egg cell, three antipodals, two polar nuclei
3. Three synergids, one egg cell, two antipodals, two polar nuclei

4. Two synergids, one egg cell, three antipodals, one polar nucleus

Options 1. 1

2. 2

3. 3

4. 4

Q.11 Match List-I with List-II

Hormones	Locations/Functions
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(A) hCG	(I) Released in the later phase of pregnancy
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(B) LH Surge	(II) Uterine contractions
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(C) Oxytocin	(III) Placenta
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(D) Relaxin	(IV) Ovulation
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Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.12 Match List-I with List-II

List-I	List-II
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Drugs	Sources/features
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(A) Alcohol	(I) Sedative and painkiller
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(B) Crack	(II) Diacetylmorphine
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(C) Morphine	(III) Liver cirrhosis
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(D) Smack	(IV) <i>Erythroxylum coca</i>
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Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.13 In 1997, the first transgenic cow, Rosie, produced human protein-enriched milk. The milk contained the _____ and was nutritionally a more balanced product for human babies than natural cow-milk.

1. Human beta-lactalbumin

2. Bovine alpha-lactalbumin

3. Human alpha-lactalbumin

4. Human gamma-lactalbumin

Options 1. 1

2. 2

3. 3

4. 4

Q.14 Arrange the following steps in the process of translation in correct sequence?

(A) Binding of release factor to stop codon

(B) Movement of ribosome from codon to codon along mRNA

(C) Aminoacylation of tRNA

(D) Binding of ribosome to mRNA at AUG

Choose the correct answer from the options given below:

1. (D), (B), (A), (C)

2. (A), (D), (B), (C)

3. (B), (A), (D), (C)
4. (C), (D), (B), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.15 A woman has given birth to fraternal twin daughters. How many ovum/ova would have been released by the ovary?

1. One
2. Two
3. Three
4. Four

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Which one of the following facts is not true about genetic equilibrium?

1. Allelic frequencies in a population change from generation to generation
2. The sum total of all allelic frequencies is 1
3. Allelic frequencies in a population are stable
4. Mutation and natural selection can change genetic equilibrium

Options 1. 1

2. 2
3. 3
4. 4

Q.17 Which one of the given statements defines the term 'parturition'?

1. The average duration of human pregnancy
2. The process of delivery of foetus
3. Production of milk by mammary glands
4. The milk produced during initial few days of lactation

Options 1. 1

2. 2
3. 3
4. 4

Q.18 Match List-I with List-II

Process/terms related to ecosystem

Explanations

(A) Decomposition

(I) Highly resistant to microbial actions

(B) Humus

(II) Oxygen dependent process

(C) Stratification

(III) Break down of detritus

(D) Fragmentation

(IV) Distribution of species

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.19 Identify the features developed by flowering plants to discourage self-pollination:

- (A) Self incompatibility
(B) Synchronized pollen release and stigma receptivity
(C) Different positions of anther and stigma in a flower

(D) Producing bisexual flowers

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (B) and (D) only
3. (A) and (C) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.20 What is nucleosome?

1. Positively charged DNA wrapped around the negatively charged histone octamer
2. 200 bases of DNA strand
3. Negatively charged DNA wrapped around the positively charged histone octamer
4. Densely packed heterochromatin

Options 1. 1

2. 2
3. 3
4. 4

Q.21 A student got data showing productivity of three different ecosystems, but the unit was missing, Choose the correct unit.

1. $\text{gm}^{-2} \text{yr}^{-1}$
2. $\text{gm}^{-3} \text{yr}^{-2}$
3. $(\text{Kcal m}^{-2}) \text{yr}^{-2}$
4. $(\text{Kcal m}^{-3}) \text{yr}^{-2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Match List-I with List-II

List-I

List-II

Contraceptive methods

Examples

(A) Non-medicated IUD

(I) LNG-20

(B) Hormone releasing IUD

(II) Multiload 375

(C) Copper releasing IUD

(III) Vaults

(D) Barrier method

(IV) Lippes loop

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.23 Last year, in a rat population of 600 rats, 600 new rats were added through reproduction in the current year to make a total population of 1200 rats.

Find the birth rate of this rat population.

1. 0.5 Offspring per month
2. 2 Offsprings per year
3. 0.5 Offspring per rat per year
4. 600/1200 offspring per week

Options 1. 1

2. 2
3. 3

Q.24 Match List-I with List-II**List-I****Terms**

(A) RNA interference

(B) Genetically engineered insulin

(C) Gene therapy

(D) Biological product

List-II**Applications**

(I) ADA deficiency

(II) α -1-antitrypsin(III) *Meloidegryne incognitia*

(IV) Eli Lilly

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.25 Identify the correct sequence of events for the technique of DNA fingerprinting.

(A) Hybridization using labelled VNTR probe, followed by detection of hybridized DNA fragments by autoradiography

(B) Transfer of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon

(C) Isolation and digestion of DNA by restriction endonucleases

(D) Separation of DNA fragments by electrophoresis

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (C), (D), (B), (A)

3. (B), (A), (D), (C)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.26 Resource partitioning among the species is the mechanism to avoid:

1. Competition

2. Predation

3. Parasitism

4. Amensalism

Options 1. 1

2. 2

3. 3

4. 4

Q.27 'The Evil Quartet' of biodiversity loss is -

1. Alien species invasion, co-extinctions, over-exploitation, sacred groves

2. Habitat loss and fragmentation, over-exploitation, sacred groves, mutualism

3. Alien species invasion, co-extinctions, habitat loss and fragmentation, over-exploitation

4. Alien species invasion, endemism, habitat loss and fragmentation, over-exploitation

Options 1. 1

2. 2

3. 3

4. 4

Q.28 Which one of the following structures is not the male sex accessory duct?

1. Seminal vesicles
2. Vas deferens
3. Vasa efferentia
4. Rete testis

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Arrange the following evolutionary human species in order of their appearance on earth from the current to the oldest one.

- (A) *Homo erectus*
- (B) Neanderthal man
- (C) *Dryopethicus*
- (D) *Homo sapiens*

Choose the correct answer from the options given below:

1. (D), (B), (A), (C)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.30 In the case of co-evolved plant pollinator mutualism where one of the species got extinct, what would be the impact on the other species?

1. It will become dormant
2. It will survive
3. It will get extinct
4. It will under go metamorphosis

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Identify the correct sequence of trophic levels in a food-chain.

- (A) Man
- (B) Phytoplankton
- (C) Fishes
- (D) Zooplankton

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (B), (C), (A)
3. (B), (D), (C), (A)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Which of the following two STIs can be transmitted through contaminated surgical equipment and are not completely curable?

1. Gonorrhoea and cholera
2. Hepatitis-B and AIDS
3. AIDS and gonorrhoea
4. Syphilis and trichomoniasis

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Which one of the following tests would be recommended by the doctor to confirm typhoid ?

1. Widal test
2. CT Scan
3. Sputum test
4. Blood sugar test

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Reserpine is an active chemical of which plant?

1. *Rauwolfia vomitoria*
2. *Azadirachta indica*
3. *Carica papaya*
4. *Mangifera indica*

Options 1. 1

2. 2
3. 3
4. 4

Q.35 Identify the incorrect statement about the ethical issues related to genetic engineering.

1. Genetic modification of organisms can have unpredictable results.
2. Manipulation of living organisms by humans cannot be done without any regulation.
3. GEAC was set up by the UN, makes decisions regarding the validity of GM research.
4. Some nations are developing laws to prevent biopiracy.

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Which of the following statements is/are true with respect to the MTP (Amendment) Act, 2017 enacted by the GOI?

- (A) A pregnancy may be terminated on certain considered grounds within the first 12 weeks of pregnancy on the opinion of two government registered medical practitioners.
- (B) This act intends to reduce illegal abortions and consequent maternal mortality and morbidity.
- (C) If the continuation of pregnancy involves a risk to the life of a pregnant woman, it can be terminated by the opinion of 3 government registered medical practitioners.
- (D) If there is a substantial risk that the child born will be seriously handicapped, pregnancy can be terminated even if lasted for more than 12 weeks, but fewer than 24 weeks by the opinion of two government registered medical practitioners.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (B) and (D) only
3. (C) and (D) only
4. (B) only

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Which one of the following statements is not the salient feature of Human Genome Project?

1. The average gene consists of 3000 bases, but sizes vary greatly, with the largest known human gene being dystrophin at 2.4 million bases.
2. The total number of genes is estimated at 30,000-much lower than previous estimates of 80,000 to 1,40,000 genes.
3. Less than 9 per cent of the genome codes for proteins.
4. Repetitive sequences are stretches of DNA sequences that are repeated many times, sometimes hundred to thousand times.

Options 1. 1

2. 2
3. 3
4. 4

Q.38 Choose the correct option, which explains the term alleles.

1. The alternative forms of the same gene
2. Sequence of DNA of different genes
3. Alphabetical symbols for genes
4. Dominant and recessive alleles

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Which one of the following techniques are used for early detection of the diseases?

1. PCR and serum analysis
2. PCR and ELISA
3. ELISA and urine analysis
4. Urine and serum analysis

Options 1. 1

2. 2
3. 3
4. 4

Q.40 A student was fascinated with the beautiful mauve colored flowers of water hyacinth, he asked his teacher to allow him to take a few plants to his village pond, but teacher denied him to do so because:

1. It is a rare species.
2. It is an endangered species.
3. It has many poisonous thorns all over its body.
4. It is an invasive weed species.

Options 1. 1

2. 2
3. 3
4. 4

Q.41 DNA recombinant technology involves the use of restriction endonucleases. After having cloned the gene of interest and having optimised the conditions to induce the expression of the target protein, one has to consider producing it on a large scale. The cells harbouring cloned genes of interest may be grown on a small scale in the laboratory. In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments, which will complicate the gene cloning. In Biolistics or gene gun method cells are bombarded with high velocity micro particles of gold or tungsten coated with DNA. The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes. For example, you can ligate a foreign DNA at the BamH I site of tetracycline resistance gene in the vector pBR322. Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol.

If any protein encoding gene is expressed in a heterologous host, it is called as

1. Recombinant DNA
2. Transgenic animal
3. Recombinant protein
4. Vector

Options 1. 1

2. 2

3. 3

4. 4

Q.42 DNA recombinant technology involves the use of restriction endonucleases. After having cloned the gene of interest and having optimised the conditions to induce the expression of the target protein, one has to consider producing it on a large scale. The cells harbouring cloned genes of interest may be grown on a small scale in the laboratory. In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments, which will complicate the gene cloning. In Biolistics or gene gun method cells are bombarded with high velocity micro particles of gold or tungsten coated with DNA. The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes. For example, you can ligate a foreign DNA at the BamH I site of tetracycline resistance gene in the vector pBR322. Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol.

pBR322, which is used in recombinant DNA technology, is a -

1. Vector
2. Matrix for gel electrophoresis
3. Enzyme
4. Recombinant chromosome

Options 1. 1

2. 2

3. 3

4. 4

Q.43 DNA recombinant technology involves the use of restriction endonucleases. After having cloned the gene of interest and having optimised the conditions to induce the expression of the target protein, one has to consider producing it on a large scale. The cells harbouring cloned genes of interest may be grown on a small scale in the laboratory. In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments, which will complicate the gene cloning. In Biolistics or gene gun method cells are bombarded with high velocity micro particles of gold or tungsten coated with DNA. The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes. For example, you can ligate a foreign DNA at the BamH I site of tetracycline resistance gene in the vector pBR322. Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol.

Which one of the following enzymes works as molecular glue?

1. Exonuclease
2. Ligase
3. Endonuclease
4. Polymerase

Options 1. 1

2. 2

3. 3

4. 4

Q.44 DNA recombinant technology involves the use of restriction endonucleases. After having cloned the gene of interest and having optimised the conditions to induce the expression of the target protein, one has to consider producing it on a large scale. The cells harbouring cloned genes of interest may be grown on a small scale in the laboratory.

In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments, which will complicate the gene cloning. In Biolistics or gene gun method cells are bombarded with high velocity micro particles of gold or tungsten coated with DNA. The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes. For example, you can ligate a foreign DNA at the BamH I site of tetracycline resistance gene in the vector pBR322. Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol.

Which one of the following statements is correct with reference to biolistics?

1. Bombarding of cells with macro particles
2. Velocity of particles to be kept low
3. Particles are coated with DNA
4. Zinc and nickel particles are used

Options 1. 1

2. 2
3. 3
4. 4

Q.45 DNA recombinant technology involves the use of restriction endonucleases. After having cloned the gene of interest and having optimised the conditions to induce the expression of the target protein, one has to consider producing it on a large scale. The cells harbouring cloned genes of interest may be grown on a small scale in the laboratory. In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes. Presence of more than one recognition sites within the vector will generate several fragments, which will complicate the gene cloning. In Biolistics or gene gun method cells are bombarded with high velocity micro particles of gold or tungsten coated with DNA. The ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes. For example, you can ligate a foreign DNA at the BamH I site of tetracycline resistance gene in the vector pBR322. Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol.

Which one of the following enzymes should be used to break the bacterial cells for isolation of DNA?

1. Lysozyme
2. Cellulase
3. Chitinase
4. Carboxylase

Options 1. 1

2. 2
3. 3
4. 4

Q.46 We use microbes and microbially derived products almost every day. Bacteria called lactic acid bacteria (LAB) grow in milk to convert it into curd. The dough, which is used to make bread, is fermented by yeast called *Saccharomyces cerevisiae*. Microbes are used to produce industrial products like lactic acid, acetic acid and alcohol, which are used in a variety of processes in the industry. However, certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO₂ and H₂. These bacteria are collectively called methanogens, and one such common bacterium is *Methanobacterium*. These bacteria are commonly found in the anaerobic sludge during sewage treatment. Microbes are also used for commercial and industrial production of certain chemicals like organic acids, alcohols and enzymes.

Examples of acid producers are *Aspergillus niger* (a fungus) of citric acid. You are familiar with the commonly used antibiotic Penicillin. Do you know that Penicillin was the first antibiotic to be discovered, and it was a chance discovery? Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

Which one of the following statements is not true with respect to microbes?

1. Wine is produced by fermentation without distillation.
2. Microbes can also be used to kill harmful pests, a process called as biocontrol.
3. The puffed up appearance of dough is due to the production of CO₂ gas by bacteria.
4. Methanogens are anaerobic bacteria.

Options

1. 1

2. 2
3. 3
4. 4

Q.47 We use microbes and microbially derived products almost every day. Bacteria called lactic acid bacteria (LAB) grow in milk to convert it into curd. The dough, which is used to make bread, is fermented by yeast called *Saccharomyces cerevisiae*. Microbes are used to produce industrial products like lactic acid, acetic acid and alcohol, which are used in a variety of processes in the industry. However, certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO₂ and H₂. These bacteria are collectively called methanogens, and one such common bacterium is *Methanobacterium*. These bacteria are commonly found in the anaerobic sludge during sewage treatment. Microbes are also used for commercial and industrial production of certain chemicals like organic acids, alcohols and enzymes. Examples of acid producers are *Aspergillus niger* (a fungus) of citric acid. You are familiar with the commonly used antibiotic Penicillin. Do you know that Penicillin was the first antibiotic to be discovered, and it was a chance discovery? Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

Citric acid is commercially obtained by *Aspergillus niger*, which is a:

1. Bacterium
2. Fungus
3. Algae
4. Virus

Options

1. 1

2. 2
3. 3
4. 4

Q.48 We use microbes and microbially derived products almost every day. Bacteria called lactic acid bacteria (LAB) grow in milk to convert it into curd. The dough, which is used to make bread, is fermented by yeast called *Saccharomyces cerevisiae*. Microbes are used to produce industrial products like lactic acid, acetic acid and alcohol, which are used in a variety of processes in the industry. However, certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO₂ and H₂. These bacteria are collectively called methanogens, and one such common bacterium is *Methanobacterium*. These bacteria are commonly found in the anaerobic sludge during sewage treatment. Microbes are also used for commercial and industrial production of certain chemicals like organic acids, alcohols and enzymes. Examples of acid producers are *Aspergillus niger* (a fungus) of citric acid. You are familiar with the commonly used antibiotic Penicillin. Do you know that Penicillin was the first antibiotic to be discovered, and it was a chance discovery? Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by:

1. Uncompetitively inhibiting the enzyme responsible for synthesis of cholesterol
2. Competitively inhibiting the enzyme responsible for synthesis of cholesterol
3. Competitively inhibiting the enzyme responsible for synthesis of triglycerides
4. Competitively inhibiting the enzyme responsible for synthesis of phospholipids

Options 1. 1

2. 2

3. 3

4. 4

Q.49 We use microbes and microbially derived products almost every day. Bacteria called lactic acid bacteria (LAB) grow in milk to convert it into curd. The dough, which is used to make bread, is fermented by yeast called *Saccharomyces cerevisiae*. Microbes are used to produce industrial products like lactic acid, acetic acid and alcohol, which are used in a variety of processes in the industry. However, certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO₂ and H₂. These bacteria are collectively called methanogens, and one such common bacterium is *Methanobacterium*. These bacteria are commonly found in the anaerobic sludge during sewage treatment. Microbes are also used for commercial and industrial production of certain chemicals like organic acids, alcohols and enzymes. Examples of acid producers are *Aspergillus niger* (a fungus) of citric acid. You are familiar with the commonly used antibiotic Penicillin. Do you know that Penicillin was the first antibiotic to be discovered, and it was a chance discovery? Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

The discovery of penicillin was a chance discovery by Alexander Fleming while working on:

1. *Staphylococci*
2. *Penicillium notatum*
3. *Lactobacillus*
4. *Clostridium butylicum*

Options 1. 1

2. 2

3. 3

4. 4

Q.50 We use microbes and microbially derived products almost every day. Bacteria called lactic acid bacteria (LAB) grow in milk to convert it into curd. The dough, which is used to make bread, is fermented by yeast called *Saccharomyces cerevisiae*. Microbes are used to produce industrial products like lactic acid, acetic acid and alcohol, which are used in a variety of processes in the industry. However, certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO₂ and H₂. These bacteria are collectively called methanogens, and one such common bacterium is *Methanobacterium*. These bacteria are commonly found in the anaerobic sludge during sewage treatment. Microbes are also used for commercial and industrial production of certain chemicals like organic acids, alcohols and enzymes. Examples of acid producers are *Aspergillus niger* (a fungus) of citric acid. You are familiar with the commonly used antibiotic Penicillin. Do you know that Penicillin was the first antibiotic to be discovered, and it was a chance discovery? Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

A boy wants to set curd from milk. Suggest which of the following microbes can be useful for him?

1. *Bacillus thuringiensis*
2. *Nucleopolyhedrovirus*
3. *Rhizobium*
4. *Lactobacillus*

Options 1. 1

4.4